



**P. D. E. A's**  
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Affiliated to  
**Savitribai Phule Pune University [SPPU]**  
**CSMEP-352 A Web Technology-I**  
**laboratory**  
**T. Y. B. Sc. (Computer Science)**  
**SEMESTER V**  
**(From Academic Year 2025)**

|                      |  |                      |  |
|----------------------|--|----------------------|--|
| <b>Student Name:</b> |  |                      |  |
| <b>Roll No:</b>      |  | <b>Exam Seat No:</b> |  |
| <b>Year:</b>         |  | <b>Division:</b>     |  |

Prepared By:

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## About The WorkBook

Objectives –

1. The scope of the course.
2. Bringing uniformity in the way course is conducted across different Colleges.
3. Continuous assessment of the students.
4. Providing ready references for students while working in the lab.

How to use this book?

This book is mandatory for the completion of the laboratory course. It is a Measure of the performance of the student in the laboratory for the entire duration of the course.

Instructions to the students

- 1) Students should carry this book during practical sessions of Computer Science.
- 2) Students should maintain separate journal for the source code and outputs.
- 3) Students should read the topics mentioned in reading section of this Book before coming for practical.
- 4) Students should solve all exercises which are selected by Practical in-charge as a part of journal activity.
- 5) Students will be assessed for each exercise on a scale of 5

|   |                   |   |
|---|-------------------|---|
| 1 | Note done         | 0 |
| 2 | Incomplete        | 1 |
| 3 | Late complete     | 2 |
| 4 | Needs improvement | 3 |
| 5 | Complete          | 4 |
| 6 | Well-done         | 5 |

### Instructions to the practical in-charge

1. Explain the assignment and related concepts in around ten minutes using whiteboard if required or by demonstrating the software.
2. Choose appropriate problems to be solved by student.
3. After a student completes a specific set, the instructor has to verify the outputs and sign in the provided space after the activity.
4. Ensure that the students use good programming practices.
5. You should evaluate each assignment carried out by a student on a scale of 5 as specified above ticking appropriate box.
6. The value should also be entered on assignment completion page of respected lab course.

PHP Semester – 1  
Assignment Completion Sheet

| Sr. No. | Assignment Name                       | Marks(out of 5) | Sign |
|---------|---------------------------------------|-----------------|------|
| 1       | Assignment Using Functions            |                 |      |
| 2       | Assignment Using Strings              |                 |      |
| 3       | Assignment Using Arrays               |                 |      |
| 4       | Assignment Using DATABASE(PostgreSQL) |                 |      |
| 5       | Assignment Using Files                |                 |      |
|         | Total out of 25                       |                 |      |
|         | Total out of 05                       |                 |      |

## Assignment No. 1: To study functions

### User-defined functions

A function may be defined using syntax such as the following:

```
function function_name([argument_list...])
{
    [statements]
    [return return_value;]
}
```

Any valid PHP code may appear inside a function, even other functions and class definitions. The variables you use inside a function are, by default, not visible outside that function. In PHP3 functions must be defined, before they are referenced. No such requirement exists in PHP4.

Example 1.

| Code                                                                                                                                       | Output |
|--------------------------------------------------------------------------------------------------------------------------------------------|--------|
| <pre>&lt;?php msg("Hello");           // calling a function function msg(\$a)         // defining a function {     echo \$a; } ?&gt;</pre> | Hello  |

### Default parameters

You can give default values to more than one argument, but once you start assigning default values, you have to give them to all arguments that follow as well.

Example 2.

| Code                                                                                                                                                               | Output            |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| <pre>&lt;?php function display(\$greeting, \$message="Good Day") {     echo \$greeting;     echo "&lt;br&gt;";     echo \$message; } display("Hello"); ?&gt;</pre> | Hello<br>Good Day |

### Variable parameters

You can set up functions that can take a variable number of arguments. Variable number of arguments can be handled with these functions:

*func\_num\_args* : Returns the number of arguments passed

*func\_get\_arg* : Returns a single argument

*func\_get\_args* : Returns all arguments in an array

| Code                                                                                                                                                                                                                                                                                                          | Output                                                        |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|
| <pre> &lt;?php echo "Passing 3 arg. to xconcat&lt;br&gt;"; echo "Result is ..."; xconcat("How","are","you"); function xconcat( ) {     \$ans = "";     \$arg = func_get_args( );     for (\$i=0; \$i&lt;func_num_args( ); \$i++ )     {         \$ans.= \$arg[\$i]. " ";     }     echo \$ans; } ?&gt; </pre> | <p>Passing 3 arg. to xconcat<br/>Result is ...How are you</p> |

### Missing parameters

When using default arguments, any defaults should be on the right side of any non-default arguments, otherwise, things will not work as expected.

Example 4.

| Code                                                                                                                                                                                         | Output                                                                                   |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| <pre> &lt;?php function makecoffee (\$type = "Nescafe") {     return "Making a cup of \$type&lt;br&gt;"; } echo makecoffee (); echo makecoffee ("espresso"); ?&gt; </pre>                    | <p>Making a cup of Nescafe.<br/>Making a cup of espresso.</p>                            |
| <pre> &lt;?php function make (\$type = "acidophilus", \$flavour) {     return "Making a bowl of \$type \$flavour&lt;br&gt;"; } echo make ("raspberry");           // won't work ?&gt; </pre> | <p>Warning: Missing argument 2 in call to make().....<br/>Making a bowl of raspberry</p> |
| <pre> &lt;?php function make (\$flavour, \$type = "acidophilus") {     return "Making a bowl of \$type \$flavour&lt;br&gt;"; } echo make ("raspberry");           //it works ?&gt; </pre>    | <p>Making a bowl of acidophilus raspberry.</p>                                           |

## Variable functions

Assign a variable the name of a function, and then treat that variable as though it is the name of a function.

Example 5.

| Code                                                                                                                                                                                                                                                                                    | Output                                         |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|
| <pre>&lt;?php \$varfun='fun1'; \$varfun(); \$varfun='fun2'; \$varfun(); \$varfun='fun3'; \$varfun(); function fun1() {     echo "&lt;br&gt;Function one"; } function fun2() {     echo "&lt;br&gt;Function two"; } function fun3() {     echo "&lt;br&gt;Function three"; } ?&gt;</pre> | Function one<br>Function two<br>Function three |

## Anonymous functions

The function that does not possess any name are called anonymous functions. Such functions are created using *create\_function()* built-in function. Anonymous functions are also called as lambda functions.

Example 6.

| Code                                                                                                                 | Output |
|----------------------------------------------------------------------------------------------------------------------|--------|
| <pre>&lt;?php \$name=create_function('\$a,\$b',     '\$c = \$a + \$b; return \$c;'); echo \$name(10,20); ?&gt;</pre> | 30     |

## Printing functions

In PHP there are various methods to print data. There are following ways to send output to the browser:

1. echo construct : Print many values at once.
2. print() :Print only one value.
3. printf() :Build a formatted string.
4. print\_r(): Prints the content of arrays, objects and things, in a more-or-less human readable form.
5. var\_dump() : prints the content of arrays, objects and other things in more human redable form along with size, type, length of elements useful for debugging.

| Code                                                                                                                                    | Output                                                 |
|-----------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|
| <pre>echo : &lt;?php echo "Printy"; echo ("Printy");//valid echo "First", "Second","Third"; echo ("Hello","Word");//invalid ?&gt;</pre> | Printy<br>Printy<br>First Second Third                 |
| <pre>print(): &lt;?php If(! print("Hello World")) { die("You are not listening to me"); } ?&gt;</pre>                                   | Hello Word                                             |
| <pre>printf() : &lt;?php printf("%.2f",27.452); printf('The hex value of %d is %x.',214,214); ?&gt;</pre>                               | 27.45<br>The hex value of 214 is d6.                   |
| <pre>print_r(): &lt;?php \$a = array('name' =&gt; 'Amar', 'age'=&gt;'35'); print_r(\$a); ?&gt;</pre>                                    | Array<br>{<br>[name]=>Amar<br>[age]=>35<br>}           |
| <pre>var_dump() : &lt;?php var_dump(14)."&lt;br&gt;"; var_dump("Hello")."&lt;br&gt;"; var_dump(25,"TY1"); ?&gt;</pre>                   | Int(14)<br>String(5) "Hello"<br>Int(25)String(3) "TY1" |

### **SET A**

Q: 1) Write a script to accept two integers(Use html form having 2 textboxes).

Write a PHP script to,

- a. Find mod of the two numbers.
- b. Find the power of first number raised to the second.
- c. Find the sum of first n numbers (considering first number as n)
- d. Find the factorial of second number.

(Write separate function for each of the above operations.)

Q: 2) Write a script to accept an integer(Use html form having 1 textbox).

Write a PHP script to,

- a. Find palindrome of a number.
- b. Find the square root of a number.

### **SET B**

Q: 1) Write a PHP script for the following: Design a form to accept the marks of 5 different subjects of a student, having serial number, subject name & marks out of 100. Display the result in the tabular format which will have total, percentage and grade. Use only 3 text boxes.(Use array of form parameters)

Q. 2) : Write a PHP script having 3 textboxes. The first textbox should be for accepting name of the student, second for accepting name of college and third for accepting a proper greeting message. Write a function for accepting all the three parameters and generating a proper greeting message. If any of the parameters are not passed, generate the proper greeting message.

(Use the concept of missing parameters)

Signature of the instructor : \_\_\_\_\_ Date : \_\_\_\_\_

Assignment Evaluation

|              |                          |                     |                          |             |                          |
|--------------|--------------------------|---------------------|--------------------------|-------------|--------------------------|
| 0:Not Done   | <input type="checkbox"/> | 2:Late Complete     | <input type="checkbox"/> | 4:Complete  | <input type="checkbox"/> |
| 1:Incomplete | <input type="checkbox"/> | 3:Needs Improvement | <input type="checkbox"/> | 5:Well Done | <input type="checkbox"/> |

## Assignment No. 2: To study Strings

### Strings in PHP

- Single quoted string (few escape characters supported, variable interpolation not possible)
- Double quoted string (many escape characters supported, variable interpolation possible)
- Heredoc

There are functions to print the string, namely print, printf, echo.

The print statement can print only single value, whereas echo and printf can print multiple values.

printf requires format specifiers. If echo statement is used like a function, then only one value can be printed.

#### Comparing Strings

Example 1.

| Code                                                                                                                                                                                                                                                                                                                                                                                                                                     | Output                                                     |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------|
| <pre>&lt;?php \$a='amit'; \$b='anil'; if(\$a==\$b)           //using operator echo "Both strings are equal&lt;br&gt;"; else     echo "Both strings are not equal&lt;br&gt;"; if(strcmp(\$a,\$b)&gt;0)    //using function {     echo "String2 sorts before String1"; } elseif(strcmp(\$a,\$b)==0) {     echo "both are equal"; } elseif(strcmp(\$a,\$b)&lt;0) // negative value {     echo "String1 sorts before String2"; }??&gt;</pre> | Both strings are not equal<br>String1 sorts before String2 |
| <pre>&lt;?php \$a=34; \$b='34'; if(\$a==\$b)           //using operator     echo "Both strings are equal&lt;br&gt;"; else     echo "Both strings are not equal&lt;br&gt;"; ??&gt;</pre>                                                                                                                                                                                                                                                  | Both strings are not equal                                 |

## Other string comparison functions

|                  |                                             |
|------------------|---------------------------------------------|
| strcasecmp( )    | : case in-sensitive string comparison       |
| strnatcmp( )     | : string comparison using a “natural order” |
| algorithm        |                                             |
| strnatcasecmp( ) | : case in-sensitive version of strnatcmp( ) |

substr\_compare() : Comparison of two strings from an offset, up to the length characters.  
The substr\_compare() compares 'main\_str' from position 'offset' with 'str' upto 'length' characters.  
It returns < 0 if 'main\_str' from position 'offset' is less than 'str', >0 if it is greater than 'str', and 0 if they are equal.

## String manipulation & searching string

Example 2.

| Code                                                                                                                                                                                                                                                                                                                                     | Output                                                                              |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| <pre>&lt;?php \$small="India"; \$big="India is my country"; \$str=substr(\$big,6,5); echo "&lt;br&gt;\$str"; \$cnt = substr_count(\$big,"i");</pre>                                                                                                                                                                                      | is my<br><br>There are 2 i's in India is my country<br><br>is found at 6 position   |
| <pre>echo "&lt;br&gt;There are".\$cnt." i's in \$big"; \$pos=strpos(\$big,"is"); echo "&lt;br&gt;is found at \$pos position"; \$replace=substr_replace(\$big,"Bharat",0,5); echo "&lt;br&gt;before replacement-&gt;\$big"; echo "&lt;br&gt;after replacement -&gt;\$replace"; ?&gt;</pre>                                                | before replacement->India is my country<br>after replacement ->Bharat is my country |
| <pre>&lt;?php \$n='This is a Test string'; echo substr_count(\$n, 'is'); ?&gt;</pre>                                                                                                                                                                                                                                                     | 2                                                                                   |
| <pre>&lt;?php echo substr_compare("abcde", "bc", 1, 2); echo substr_compare("abcde", "de", -2, 2); echo substr_compare("abcde", "bcg", 1, 2); echo substr_compare("abcde", "BC", 1, 2, true); echo substr_compare("abcde", "bc", 1, 3); echo substr_compare("abcde", "cd", 1, 2); echo substr_compare("abcde", "abc", 5, 1); ?&gt;</pre> | 0<br>0<br>0<br>0<br>1<br>-1<br>warning                                              |
| <pre>?php cho strrev("Hello World"); ?&gt;</pre>                                                                                                                                                                                                                                                                                         | dlroW olleH                                                                         |

|                                                                                                                                                                                                   |                                                                 |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|
| <pre>&lt;?php \$string = "Hello"; \$x = 5; \$result = str_repeat(\$string, \$x); echo \$result; ?&gt;</pre>                                                                                       | <p>Hello Hello Hello Hello Hello</p>                            |
| <pre>&lt;?php \$input = "PHP"; echo str_pad(\$input, 10); echo str_pad(\$input, 10, "=", STR_PAD_LEFT); echo str_pad(\$input, 10, "=", STR_PAD_BOTH); echo str_pad(\$input, 10, "="); ?&gt;</pre> | <p> "PHP"<br/> "=====PHP"<br/> "===PHP==="<br/> "PHP=====" </p> |

## Regular Expressions

Two types of regular expressions

POSIX – style

PERL – compatible

### Purpose of using regular expressions

Matching

Substituting

Splitting

Example 3.

| Code                                                                                                                                                                                                                                                                                                                                                                                                                | Output                                                                                                                                                                                                                            |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <pre>&lt;?php \$big=&lt;&lt;&lt; paragraph India is my country. I am proud of it. I live in Maharashtra. paragraph; echo "&lt;br&gt;"; \$found=preg_match('/am/i',\$big); if(\$found)     echo "&lt;br&gt;am found in \"\$big\""; \$replace=preg_replace('/India/','Bharat',\$big); echo "&lt;br&gt;\$replace"; \$split=preg_split('/ /',\$big); foreach(\$split as \$elem) { echo "&lt;br&gt;\$elem";} ?&gt;</pre> | <p> am found in \$big<br/> Bharat is my country. I am proud<br/> of it. I live in Maharashtra.<br/> India<br/> is<br/> my<br/> country.<br/> I<br/> am<br/> proud<br/> of<br/> it.<br/> I<br/> live<br/> in<br/> Maharashtra </p> |

### **SET A**

- Q. 1) Write a PHP script for the following: Design a form to accept a string.
- Write a function to find the length of given string without using built in functions.
  - Write a function to count the total number of vowels i.e. (a,e,i,o,u) ) from the string.
  - Convert the given string to lowercase and then to Title case.
  - Pad the given string with "\*" from left and right both the sides.
  - Remove the leading whitespaces from the given string.
  - Find the reverse of given string.(use built-in functions)
- (Use radio buttons and the concept of function. Use 'include' construct or 'require' stmt.)

- Q. 2) Write a PHP script for the following: Design a form to accept two strings from the user.
- Find whether the small string appears at the start of the large string.
  - Find the position of the small string in the big string.
  - Compare both the string for first n characters, also the comparison should not be case sensitive.

### **SET B**

Q. 1) : Write a PHP script for the following: Design a form having a text box and a drop down list containing any 3 separators(e.g. #, |, %, @, ! or comma) accept a strings from the user and also a separator.

- Split the string into separate words using the given separator.
- Replace all the occurrences of separator in the given string with some other separator.
- Find the last word in the given string(Use **strrstr()** function).

Q: 2) Write a PHP script to accept 2 strings from the user, the first string should be a sentence and second can be a word.

- Delete a small part from first string after accepting position and number of characters to remove.
  - Insert the given small string in the given big string at specified position without removing any characters from the big string.
  - Replace some characters/ words from given big string with the given small string at specified position.
  - Replace all the characters from the big string with the small string after a specified position.
- (Use **substr\_replace()** function)

### **SET C**

Q: 1) Using regular expressions check for the validity of entered email-id.

- The @ symbol should not appear more than once.
- The dot (.) can appear at the most once before @ and at the most twice or at least once after @ symbol.
- The substring before @ should not begin with a digit or underscore or dot or @ or any other special character. (Use **explode** and **ereg** function.)

Signature of the instructor : \_\_\_\_\_ Date : \_\_\_\_\_

Assignment Evaluation

|              |                      |                     |                      |             |                      |
|--------------|----------------------|---------------------|----------------------|-------------|----------------------|
| 0:Not Done   | <input type="text"/> | 2:Late Complete     | <input type="text"/> | 4:Complete  | <input type="text"/> |
| 1:Incomplete | <input type="text"/> | 3:Needs Improvement | <input type="text"/> | 5:Well Done | <input type="text"/> |

## ASSIGNMENT NO. 3 : TO STUDY ARRAYS :

An array is a collection of data values. Array is organized as an ordered collection of (key,value) pairs.

In PHP there are two kinds of arrays:

Indexed array: An array with a numeric index starting with 0.

For example, Initializing an indexed array,

```
$numbers[0]=100;
```

```
$numbers[1]=200;
```

```
$numbers[2]=300;
```

Associative array: An array which have strings as keys which are used to access the values.

Initializing an Associative array,

```
$numbers[ 'one' ]=100;
```

```
$numbers[ 'two' ]=200;
```

```
$numbers[ 'three' ]=300;
```

### **Creating Arrays**

Arrays can be created in multiple ways

1. Using simple assignment to initialize an array.

```
Eg : $addresses[0] = "spam@cyberpromo.net";
```

```
$addresses[1] = "abuse@example.com";
```

```
$addresses[2] = root@example.com
```

**OR**

```
$price['gasket'] = 15.29;
```

```
$price['wheel'] = 75.25;
```

```
$price['tire'] = 50.00
```

2. An easier way to initialize an array is to use the array() construct, which builds an array from its arguments. This builds an indexed array, and the index values (starting at 0) are created automatically:

```
$addresses = array("spam@cyberpromo.net", "abuse@example.com",  
"root@example.com");
```

To create an associative array with array(), use the => symbol to separate indices (keys) from values:

```
$price = array( 'gasket' => 15.29, 'wheel' => 75.25, 'tire' => 50.00 );
```

3. To construct an empty array, pass no arguments to array(): \$addresses = array();

Some Array Functions Are:

| Name                    | Use                                                                                                                  | Example                                                                                                                                                                                                                                                                         |
|-------------------------|----------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Array()                 | This construct is used to initialize an array.                                                                       | <pre>\$numbers=array(100,200,300); \$cities=array( 'Capital of Nation'=&gt;'Delhi', 'Capital of state'=&gt;'Mumbai', 'My city'=&gt;'Nashik');</pre>                                                                                                                             |
| List()                  | This function is used to copy values from array to the variables.                                                    | <pre>\$cities=array( 'Capital of Nation'=&gt;'Delhi', 'Capital of state'=&gt;'Mumbai', 'My city'=&gt;'Nashik');</pre>                                                                                                                                                           |
| Count(), sizeof()       | The count() and sizeof() functions are identical in use and effect. They return the number of elements in the array. | <pre>\$family = array("Fred", "Wilma", "Pebbles"); \$size = count(\$family); // \$size is 3</pre>                                                                                                                                                                               |
| array_splice()          | This function is used to remove or insert elements in array                                                          | <pre>\$student=array(11,12,13,14,15,16); \$new_student=array_splice(\$student,2,3); /* starting from index(2) and length =3 \$new_student1=array_splice(\$student,2); /* here length is not mentioned */ Output : \$new_student=(13,14,15); \$new_student1=(13,14,15,16);</pre> |
| array_key_exists()<br>) | This function is used to check if an element exist in the array.                                                     | <pre>\$cities=array( 'Capital of Nation'=&gt;'Delhi', 'Capital of state'=&gt;'Mumbai', 'My city'=&gt;'Nashik'); If (array_key_exists('Capital of State',\$cities)) { Echo "key found!\n"; } Output :Key found!</pre>                                                            |
| extract()               | This function automatically creates local variables from the array                                                   | <pre>\$student = array('roll'=&gt;11, 'name'=&gt;'A','class'=&gt;'TYBSC'); Extract(\$student); By this, the variables are created like this : \$roll = 11; \$name='A'; \$class='TYBSc';</pre>                                                                                   |
| array_filter()          | To identify a subset of an array based on its values, use the array_filter() function                                | <pre>\$callback = function isOdd (\$element) { return \$element % 2; }; \$numbers = array(9, 23, 24, 27); \$odds = array_filter(\$numbers, \$callback); // \$odds is array(0 =&gt; 9, 1 =&gt; 23, 3 =&gt; 27)</pre>                                                             |
| shuffle()               | To traverse the elements in an array in random order, use the shuffle() function                                     | <pre>\$weekdays = array("Monday", "Tuesday", "Wednesday", "Thursday", "Friday"); shuffle(\$weekdays); print_r(\$days); Array( [0] =&gt; Tuesday [1] =&gt; Thursday [2] =&gt; Monday [3] =&gt; Friday [4] =&gt; Wednesday )</pre>                                                |

| Name          | Use                                                                                                                                                                                            | Example                                                                                                                                                                                                                                                                                                                             |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Array_slice() | <p>To extract a slice of the array.</p> <p>The array_slice() will reorder and reset the numeric array indices by default. You can change this behavior by setting “preserve_keys” to True.</p> | <pre>&lt;?php \$input= array("a","b","c","d","e"); //Note the differences in the array keys print_r(array_slice(\$input,2,2)); print_r(array_slice(\$input,2,2,true)); ?&gt;</pre> <p>Output:</p> <pre>Array{ [0]=&gt;c [1]=&gt;d } Array{ [2]=&gt;c [3]=&gt;d }</pre>                                                              |
| array_chunk() | <p>This function returns a multidimensional numerically indexed array , starting with zero, with each dimension containing size elements.</p>                                                  | <pre>&lt;?php \$input_array= array("a","b","c","d","e"); print_r(array_chunk(\$input_array,2)); ?&gt;</pre> <p>Output:</p> <pre>Array {     [0]=&gt;Array     {         [0]=&gt;a         [1]=&gt;b     }     [1]=&gt;Array     {         [0]=&gt;c         [1]=&gt;d     }     [2]=&gt;Array     {         [0]=&gt;e     } }</pre> |

**Set A**

1. Create your array of 30 high temperatures, approximating the weather for a spring month, then find the average high temp, the five warmest high temps and the five coolest high temps. Display the result on the browser.  
Hint: a) Use array\_slice b) the HTML character entity for the degree sign is & deg;.
2. Write a menu driven program to perform the following stack and queue related operations:[Hint: use Array\_push(), Array\_pop(), Array\_shift(), Array\_unshift() ]
  - a) Insert an element in stack
  - b) Delete an element from stack
  - c) Display the contents of stack
  - d) Insert an element in queue
  - e) Delete an element from queue
  - f) Display the contents of queue

**Set B**

1. Write a PHP script that inserts a new item in an array at any position.  
(hint : use array\_splice())
2. Define an array. Find the elements from the array that matches the given value using appropriate search function.

**Set C**

1. Write a PHP script to sort the following associative array :  
array("Sophia"=>"31","Jacob"=>"41","William"=>"39","Ramesh"=>"40") in
  - a) ascending order sort by value
  - b) ascending order sort by Key
  - c) descending order sorting by Value
  - d) descending order sorting by Key
2. Write a menu driven program to perform the following operations on associative arrays:
  - a) Split an array into chunks
  - b) Sort the array by values without changing the keys.
  - c) Filter the odd elements from an array.
  - d) Merge the given arrays.
  - e) Find the intersection of two arrays.
  - f) Find the union of two arrays.
  - g) Find set difference of two arrays.

**Assignment Evaluation**

|                       |  |                |  |                   |  |
|-----------------------|--|----------------|--|-------------------|--|
| 0 : Not done          |  | 1 : Incomplete |  | 2 : Late Complete |  |
| 3 : Needs Improvement |  | 4 : Complete   |  | 5 : Well done     |  |

## Assignment 4: Database (PHP-PostgreSQL)

### Accessing Databases (PostgreSQL)

PostgreSQL supports a wide variety of built-in data types and it also provides an option to the users to add new data types to PostgreSQL, using the CREATE TYPE command.

Table lists the data types officially supported by PostgreSQL. Most datatypes supported by PostgreSQL are directly derived from SQL standards.

The following table contains PostgreSQL supported data types for your ready reference

| Category            | Data type                         | Description                                                    |
|---------------------|-----------------------------------|----------------------------------------------------------------|
| Boolean             | boolean, bool                     | A single true or false value.                                  |
| Binary types        | bit(n)                            | An n-length bit string (exactly n) binary bits)                |
|                     | bit varying(n), varbit(n)         | A variable n-length bit string (upto) binary nbits)            |
| Character Types     | character(n)                      | A fixed n-length character string                              |
|                     | char(n)                           | A fixed n-length character string                              |
|                     | character varying(n)              |                                                                |
|                     | varchar (n)                       |                                                                |
|                     | text                              | A variable length character string of unlimited length         |
| Numeric types       | smallint, int2                    | A signed 2-byte integer                                        |
|                     | integer, int, int4                | A signed, fixed precision 4-byte number                        |
|                     | bigint, int8                      | A signed 8-byte integer, up to 18 digits in length             |
|                     | real, float4                      | A 4-byte floating point number                                 |
|                     | float8, float                     | An 8-byte floating point number                                |
|                     | numeric(p,s)                      | An exact numeric type with arbitrary precision p, and scale s. |
| Currency            | money                             | A fixed precision, U.S style currency                          |
|                     | serial                            | An auto-incrementing 4-byte integer                            |
| Date and time types | date                              | The calendar date(day, month and year)                         |
|                     | time                              | The time of day                                                |
|                     | time with time zone               | the time of day, including time zone information               |
|                     | timestamp(includes time Interval) | An arbitrarily specified length                                |

## Functions used for PostgreSQL database manipulation

| S. No. | API & Description                                                                                                                                                                                                                                                                                                                                                                                  |
|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1      | <p><b>resource pg_connect ( string \$connection_string [, int \$connect_type ] )</b></p> <p>This opens a connection to a PostgreSQL database specified by the connection_string.</p> <p>If PGSQL_CONNECT_FORCE_NEW is passed as connect_type, then a new connection is created in case of a second call to pg_connect(), even if the connection_string is identical to an existing connection.</p> |
| 2      | <p><b>bool pg_connection_reset ( resource \$connection )</b></p> <p>This routine resets the connection. It is useful for error recovery. Returns TRUE on success or FALSE on failure.</p>                                                                                                                                                                                                          |
| 3      | <p><b>int pg_connection_status ( resource \$connection )</b></p> <p>This routine returns the status of the specified connection. Returns PGSQL_CONNECTION_OK or PGSQL_CONNECTION_BAD.</p>                                                                                                                                                                                                          |
| 4      | <p><b>string pg_dbname ([ resource \$connection ] )</b></p> <p>This routine returns the name of the database that the given PostgreSQL connection resource.</p>                                                                                                                                                                                                                                    |
| 5      | <p><b>resource pg_prepare ([ resource \$connection ], string \$stmtname, string \$query )</b></p> <p>This submits a request to create a prepared statement with the given parameters and waits for completion.</p>                                                                                                                                                                                 |
| 6      | <p><b>resource pg_execute ([ resource \$connection ], string \$stmtname, array \$params )</b></p> <p>This routine sends a request to execute a prepared statement with given parameters and waits for the result.</p>                                                                                                                                                                              |

|    |                                                                                                                                                                                      |
|----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7  | <b>resource pg_query ([ resource \$connection ], string \$query )</b><br><br>This routine executes the query on the specified database connection.                                   |
| 8  | <b>array pg_fetch_row ( resource \$result [, int \$row ] )</b><br><br>This routine fetches one row of data from the result associated with the specified result resource.            |
| 9  | <b>array pg_fetch_all ( resource \$result )</b><br><br>This routine returns an array that contains all rows (records) in the result resource.                                        |
| 10 | <b>int pg_affected_rows ( resource \$result )</b><br><br>This routine returns the number of rows affected by INSERT, UPDATE, and DELETE queries.                                     |
| 11 | <b>int pg_num_rows ( resource \$result )</b><br><br>This routine returns the number of rows in a PostgreSQL result resource for example number of rows returned by SELECT statement. |
| 12 | <b>bool pg_close ([ resource \$connection ] )</b><br><br>This routine closes the non-persistent connection to a PostgreSQL database associated with the given connection resource.   |
| 13 | <b>string pg_last_error ([ resource \$connection ] )</b><br><br>This routine returns the last error message for a given connection.                                                  |
| 14 | <b>string pg_escape_literal ([ resource \$connection ], string \$data )</b><br><br>This routine escapes a literal for insertion into a text field.                                   |
| 15 | <b>string pg_escape_string ([ resource \$connection ], string \$data )</b><br><br>This routine escapes a string for querying the database.                                           |

## Connecting to Database

**pg\_connect ( )** — is used to open a PostgreSQL connection.

**Syntax :**

**resource pg\_connect ( string \$connection\_string [, int \$connect\_type ] )**

**pg\_connect()** opens a connection to a PostgreSQL database specified by the `connection_string`.

If a second call is made to **pg\_connect()** with the same `connection_string` as an existing connection, the existing connection will be returned unless you pass `PGSQL_CONNECT_FORCE_NEW` as `connect_type`

The following PHP code shows how to connect to an existing database on a local machine and finally a database connection object will be returned.

```
$db=pg_connect("dbname=sonal");
```

```
$db=pg_connect("host=localhost port= 5432 dbname=sonal");
```

```
$db=pg_connect("host=localhost port=5432 dbname=sonal  
user=postgres password=redhat");
```

```
<?php  
$host      = "host = 127.0.0.1";  
$port      = "port = 5432";  
$dbname    = "dbname = testdb";  
$credentials = "user = postgres password=pass123";  
$db = pg_connect( "$host $port $dbname $credentials" );  
if(!$db) {  
    echo "Error : Unable to open database\n";  
} else {  
    echo "Opened database successfully\n";  
}
```

```
?>
```

Now, let us run the above given program to open our database **testdb**: if the database is successfully opened, then it will give the following message –

```
Opened database successfully
```

## Closing Connection:

**pg\_close()**- function closes a PostgreSQL connection

**pg\_close** — Closes a PostgreSQL connection

### syntax :

```
bool pg_close ([ resource $connection ] )
```

**pg\_close()** closes the non-persistent connection to a PostgreSQL database associated with the given connection resource.

## Execute AQuery :

**pg\_query** — to Execute a query

### syntax :

```
resource pg_query ([ resource $connection ], string $query )
```

**pg\_query()** executes the query on the specified database connection.

## Createa Table

The following PHP program will be used to create a table in a previously created database –

```
<?php
$host      = "host = 127.0.0.1";
$port      = "port = 5432";
$dbname    = "dbname = testdb";
$credentials = "user = postgres password=pass123";

$db = pg_connect( "$host $port $dbname $credentials" );
if(!$db) {
```

```

        echo "Error : Unable to open database\n";
    } else {
        echo "Opened database successfully\n";
    }

    $sql =<<<EOF
    CREATE TABLE COMPANY
    (ID INT PRIMARY KEY   NOT NULL,
    NAME      TEXT   NOT NULL,
    AGE      INT   NOT NULL,
    ADDRESS   CHAR(50),
    SALARY    REAL);
EOF;
    $ret = pg_query($db, $sql);
    if(!$ret) {
        echo pg_last_error($db);
    } else {
        echo "Table created successfully\n";
    }
    pg_close($db);
?>

```

When the above given program is executed, it will create COMPANY table in your **testdb** and it will display the following messages –

```

Opened database successfully
Table created successfully

```

## INSERTOperation

The following PHP program shows how we can create records in our COMPANY table created in above example –

```

<?php
    $host      = "host=127.0.0.1";
    $port      = "port=5432";
    $dbname    = "dbname = testdb";

```

```

$credentials = "user = postgres password=pass123";

$db = pg_connect( "$host $port $dbname $credentials" );
if(!$db) {
    echo "Error : Unable to open database\n";
} else {
    echo "Opened database successfully\n";
}

$sql =<<<EOF
INSERT INTO COMPANY (ID,NAME,AGE,ADDRESS,SALARY)
VALUES (1, 'Paul', 32, 'California', 20000.00 );

INSERT INTO COMPANY (ID,NAME,AGE,ADDRESS,SALARY)
VALUES (2, 'Allen', 25, 'Texas', 15000.00 );

INSERT INTO COMPANY (ID,NAME,AGE,ADDRESS,SALARY)
VALUES (3, 'Teddy', 23, 'Norway', 20000.00 );

INSERT INTO COMPANY (ID,NAME,AGE,ADDRESS,SALARY)
VALUES (4, 'Mark', 25, 'Rich-Mond ', 65000.00 );
EOF;

$ret = pg_query($db, $sql);
if(!$ret) {
    echo pg_last_error($db);
} else {
    echo "Records created successfully\n";
}
pg_close($db);

```

When the above given program is executed, it will create the given records in COMPANY table and will display the following two lines –

```
Opened database successfully
Records created successfully
```

## **pg\_fetch\_row**

`pg_fetch_row` — Get a row as an enumerated array

Description

array **pg\_fetch\_row** ( resource \$result [, int \$row ] )

**pg\_fetch\_row()** fetches one row of data from the result associated with the specified result resource.

**Note:** This function sets NULL fields to the PHP NULL value.

Parameters\_

**result**

PostgreSQL query result resource, returned by `pg_query()`, `pg_query_params()` or `pg_execute()` (among others).

row

Row number in result to fetch. Rows are numbered from 0 upwards. If omitted or NULL, the next row is fetched.

Return Values

An array, indexed from 0 upwards, with each value represented as a string. Database *NULL* values are returned as NULL.

FALSE is returned if row exceeds the number of rows in the set, there are no more rows, or on any other error.

## SELECT Operation

The following PHP program shows how we can fetch and display records from our COMPANY table created in above example –

```
<?php
$host      = "host = 127.0.0.1";
$port      = "port = 5432";
$dbname    = "dbname = testdb";
$credentials = "user = postgres password=pass123";

$db = pg_connect( "$host $port $dbname $credentials" );
if(!$db) {
    echo "Error : Unable to open database\n";
} else {
    echo "Opened database successfully\n";
}

$sql = <<<EOF
    SELECT * from COMPANY;
EOF;

$ret = pg_query($db, $sql);
if(!$ret) {
    echo pg_last_error($db);
    exit;
}

while($row = pg_fetch_row($ret)) {
    echo "ID = ". $row[0] . "\n";
    echo "NAME = ". $row[1] . "\n";
    echo "ADDRESS = ". $row[2] . "\n";
    echo "SALARY = ". $row[4] . "\n\n";
}

echo "Operation done successfully\n";

pg_close($db);
```

?>

When the above given program is executed, it will produce the following result. Keep a note that fields are returned in the sequence they were used while creating table.

Opened database successfully

ID = 1

NAME = Paul

ADDRESS = California

SALARY = 20000

ID = 2

NAME = Allen

ADDRESS = Texas

SALARY = 15000

ID = 3

NAME = Teddy

ADDRESS = Norway

SALARY = 20000

ID = 4

NAME = Mark

ADDRESS = Rich-Mond

SALARY = 65000

Operation done successfully

## UPDATEOperation

The following PHP code shows how we can use the UPDATE statement to update any record and then fetch and display updated records from our COMPANY table –

```
<?php
```

```
$host      = "host=127.0.0.1";
```

```
$port      = "port=5432";
```

```
$dbname    = "dbname = testdb";
```

```
$credentials = "user = postgres password=pass123";
```

```
$db = pg_connect( "$host $port $dbname $credentials" );
```

```
if(!$db) {
```

```
    echo "Error : Unable to open database\n";
```

```
} else {
```

```
    echo "Opened database successfully\n";
```

```

}

$sql =<<<EOF
    UPDATE COMPANY set SALARY = 25000.00 where ID=1;
EOF;

$ret = pg_query($db, $sql);
if(!$ret) {
    echo pg_last_error($db);
    exit;
} else {
    echo "Record updated successfully\n";
}

$sql =<<<EOF
    SELECT * from COMPANY;
EOF;

$ret = pg_query($db, $sql);
if(!$ret) {
    echo pg_last_error($db);
    exit;
}
while($row = pg_fetch_row($ret)) {
    echo "ID = ". $row[0] . "\n";
    echo "NAME = ". $row[1] . "\n";
    echo "ADDRESS = ". $row[2] . "\n";
    echo "SALARY = ". $row[4] . "\n\n";
}
echo "Operation done successfully\n";
pg_close($db);
?>

```

When the above given program is executed, it will produce the following result –

Opened database successfully

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Record updated successfully

ID = 2

NAME = Allen

ADDRESS = 25

SALARY = 15000

ID = 3

NAME = Teddy

ADDRESS = 23

SALARY = 20000

ID = 4

NAME = Mark

ADDRESS = 25

SALARY = 65000

ID = 1

NAME = Paul

ADDRESS = 32

SALARY = 25000

Operation done successfully

## DELETEOperation

The following PHP code shows how we can use the DELETE statement to delete any record and then fetch and display the remaining records from our COMPANY table –

```
<?php
$host      = "host = 127.0.0.1";
$port      = "port = 5432";
$dbname    = "dbname = testdb";
$credentials = "user = postgres password=pass123";

$db = pg_connect( "$host $port $dbname $credentials" );
if(!$db) {
    echo "Error : Unable to open database\n";
} else {
    echo "Opened database successfully\n";
}

$sql =<<<EOF
DELETE from COMPANY where ID=2;
```

```

EOF;

$ret = pg_query($db, $sql);
if(!$ret) {
    echo pg_last_error($db);
    exit;
} else {
    echo "Record deleted successfully\n";
}

$sql =<<<EOF
    SELECT * from COMPANY;
EOF;

$ret = pg_query($db, $sql);
if(!$ret) {
    echo pg_last_error($db);
    exit;
}
while($row = pg_fetch_row($ret)) {
    echo "ID = ". $row[0] . "\n";
    echo "NAME = ". $row[1] . "\n";
    echo "ADDRESS = ". $row[2] . "\n";
    echo "SALARY = ". $row[4] . "\n\n";
}
echo "Operation done successfully\n";
pg_close($db);
?>

```

When the above given program is executed, it will produce the following result –

```

Opened database successfully
Record deleted successfully
ID = 3
NAME = Teddy
ADDRESS = 23
SALARY = 20000

```

ID = 4  
NAME = Mark  
ADDRESS = 25  
SALARY = 65000

ID = 1  
NAME = Paul  
ADDRESS = 32  
SALARY = 25000

Operation done successfully

### **SetA**

Q1. Consider the following entities and their relationships

Event (eno , title , date )

Committee ( cno , name, head , from\_time ,to\_time , status)

Event and Committee have many to many relationship. Write a script to accept title of event and modify status committee as working.

### **SetB**

Q1. Considerer the following entities and their relationships

Student (Stud\_id,name,class)

Competition (c\_no,c\_name,type)

Relationship between student and competition is many-many with attribute rank and year. Create a RDB in 3NF for the above and solve the following. Using above database write a script in PHP to accept a competition name from user and display information of student who has secured 1st rank in that competition.

### **SetC**

Q 2. Property (pno, description, area)

- Owner (oname, address, phone)
- An owner can have one or more properties, but a property belongs to exactly one owner.
- Accept owner name from the user. Write a PHP script which will display all properties which are own by that owner

**Signature of the instructor:** \_\_\_\_\_ **Date :** \_\_\_\_\_

Assignment Evaluation

|              |                      |                     |                      |             |                      |
|--------------|----------------------|---------------------|----------------------|-------------|----------------------|
| 0:Not Done   | <input type="text"/> | 2:Late Complete     | <input type="text"/> | 4:Complete  | <input type="text"/> |
| 1:Incomplete | <input type="text"/> | 3:Needs Improvement | <input type="text"/> | 5: Well Don | <input type="text"/> |

## Assignment 5: To study Files

**File :** A **file** is nothing more than an ordered sequence of bytes stored on hard disk, floppy disk CD-ROM or some other storage media. Operations on file are Opening and closing a file. Reading a file and writing into file Deleting and renaming a file navigating a file

A **file handler** is nothing more than an integer value that will be used to identify the file you wish to work with until it is closed working with files

| Function Name | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |      |         |   |                                                                            |    |                                                                                    |   |                                                                                                                                                                    |    |                                                                                                                                                                           |   |                                                                                                                         |    |                                                                                                                                |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|---------|---|----------------------------------------------------------------------------|----|------------------------------------------------------------------------------------|---|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|-------------------------------------------------------------------------------------------------------------------------|----|--------------------------------------------------------------------------------------------------------------------------------|
| fopen()       | <p>Opening and closing a file<br/>This is used to open a file ,returning a file handle associated with opened file .It can take three arguments :fname, mode and optional use_include_path<br/>Ex:-\$fp=fopen("data.txt",r);<br/>We can also open a file on remote hostList<br/>of modes used in fopen are:</p> <table><tr><th>Mode</th><th>Purpose</th></tr><tr><td>R</td><td>Open for reading only; place the file pointer at the beginning of the file</td></tr><tr><td>r+</td><td>Open for reading and writing; place the file pointer at the beginning of the file.</td></tr><tr><td>w</td><td>Open for writing only; place the file pointer at the beginning of the file and truncate the file to zero length. If the file does not exist, attempt to create it.</td></tr><tr><td>w+</td><td>Open for reading and writing; place the file pointer at the beginning of the file and truncate the file to zero length. If the file does not exist, attempt to create it.</td></tr><tr><td>A</td><td>Open for writing only; place the file pointer at the end of the file. If the file does not exist, attempt to create it.</td></tr><tr><td>a+</td><td>Open for reading and writing; place the file pointer at the end of the file. If the file does not exist, attempt to create it.</td></tr></table> | Mode | Purpose | R | Open for reading only; place the file pointer at the beginning of the file | r+ | Open for reading and writing; place the file pointer at the beginning of the file. | w | Open for writing only; place the file pointer at the beginning of the file and truncate the file to zero length. If the file does not exist, attempt to create it. | w+ | Open for reading and writing; place the file pointer at the beginning of the file and truncate the file to zero length. If the file does not exist, attempt to create it. | A | Open for writing only; place the file pointer at the end of the file. If the file does not exist, attempt to create it. | a+ | Open for reading and writing; place the file pointer at the end of the file. If the file does not exist, attempt to create it. |
| Mode          | Purpose                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |      |         |   |                                                                            |    |                                                                                    |   |                                                                                                                                                                    |    |                                                                                                                                                                           |   |                                                                                                                         |    |                                                                                                                                |
| R             | Open for reading only; place the file pointer at the beginning of the file                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |      |         |   |                                                                            |    |                                                                                    |   |                                                                                                                                                                    |    |                                                                                                                                                                           |   |                                                                                                                         |    |                                                                                                                                |
| r+            | Open for reading and writing; place the file pointer at the beginning of the file.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |      |         |   |                                                                            |    |                                                                                    |   |                                                                                                                                                                    |    |                                                                                                                                                                           |   |                                                                                                                         |    |                                                                                                                                |
| w             | Open for writing only; place the file pointer at the beginning of the file and truncate the file to zero length. If the file does not exist, attempt to create it.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |      |         |   |                                                                            |    |                                                                                    |   |                                                                                                                                                                    |    |                                                                                                                                                                           |   |                                                                                                                         |    |                                                                                                                                |
| w+            | Open for reading and writing; place the file pointer at the beginning of the file and truncate the file to zero length. If the file does not exist, attempt to create it.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |      |         |   |                                                                            |    |                                                                                    |   |                                                                                                                                                                    |    |                                                                                                                                                                           |   |                                                                                                                         |    |                                                                                                                                |
| A             | Open for writing only; place the file pointer at the end of the file. If the file does not exist, attempt to create it.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |      |         |   |                                                                            |    |                                                                                    |   |                                                                                                                                                                    |    |                                                                                                                                                                           |   |                                                                                                                         |    |                                                                                                                                |
| a+            | Open for reading and writing; place the file pointer at the end of the file. If the file does not exist, attempt to create it.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |      |         |   |                                                                            |    |                                                                                    |   |                                                                                                                                                                    |    |                                                                                                                                                                           |   |                                                                                                                         |    |                                                                                                                                |
| fclose()      | <p>This is used to close file, using its associated file handle as a single argument<br/>Ex:- fclose(fp);</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |      |         |   |                                                                            |    |                                                                                    |   |                                                                                                                                                                    |    |                                                                                                                                                                           |   |                                                                                                                         |    |                                                                                                                                |
| fread( )      | <p>This function is used to extract a character string from a file and takes two arguments, a file handle and a integer length<br/>Ex: fread(\$fp,10);</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |      |         |   |                                                                            |    |                                                                                    |   |                                                                                                                                                                    |    |                                                                                                                                                                           |   |                                                                                                                         |    |                                                                                                                                |
| fwrite()      | <p>This function is used to write data to a file and takes two arguments, a file handle and a string<br/>Ex: fwrite(\$fp,"HELLO");</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |      |         |   |                                                                            |    |                                                                                    |   |                                                                                                                                                                    |    |                                                                                                                                                                           |   |                                                                                                                         |    |                                                                                                                                |

|                  |                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| fgetc()          | Function can be used to read one character from file at a file. It takes a single argument ,a file handle and return just one character from the file .It returns false when it reached to end of file.                                                                                                                                                                                                                                     |
| fgets()          | This function is used to read set of characters it takes two arguments, file pointer and length.<br>It will stop reading for any one of three reasons: The specified number of bytes has been read<br>A new line is encountered<br>The end of file is reached                                                                                                                                                                               |
| fputs()          | This is simply an alias for fwrite() .                                                                                                                                                                                                                                                                                                                                                                                                      |
| file()           | This function will return entire contents of file. This function will automatically opens, reads, and close the file. It has one argument :a string containing the name of the file.It can also fetch files on remote host.                                                                                                                                                                                                                 |
| fpasssthru()     | This function reads and print the entire file to the web browser. This function takes one argument ,file handle. If youread a couple of lines from a file before calling fpassthru()<br>,then this function only print the subsequent contents of a file.                                                                                                                                                                                   |
| readfile()       | This function prints content of file without having a call to fopen().<br>It takes a filename as its argument ,reads a file and then write it to standard output returning the number of bytes read(or false upon error)                                                                                                                                                                                                                    |
| fseek()          | It takes file handle and integer offset , offset type as an arguments .It will move file position indicator associated with file pointer to a position determined by offset. By default this offset is measured in bytes from the beginning of the file.<br>The third argument is optional ,can be specified as:<br>SEEK_SET:-Beginning of file +offset<br>SEEK_CUR:-Current position +offset(default)<br>SEEK_END:-End of the file +offset |
| ftell()          | It takes file handle as an argument and returns the current offset(in bytes) of the corresponding file position indicator.                                                                                                                                                                                                                                                                                                                  |
| rewind()         | It accepts a file handle as an argument and reset the corresponding file position indicator to the beginning of file.                                                                                                                                                                                                                                                                                                                       |
| file_exists()    | It takes file name with detail path as an argument and returns true if file is there otherwise it returns false                                                                                                                                                                                                                                                                                                                             |
| file_size()      | It takes file name as an argument and returns total size of file (in bytes)                                                                                                                                                                                                                                                                                                                                                                 |
| fileatime()      | It takes filename as an argument and returns last access time for a file in a UNIX timestamp format                                                                                                                                                                                                                                                                                                                                         |
| filectime()      | It takes filename as an argument and returns the time at which the file was last changed as a UNIX timestamp format                                                                                                                                                                                                                                                                                                                         |
| filemtime()      | It takes filename as an argument and returns the time at which the file was last modified as a UNIX timestamp format                                                                                                                                                                                                                                                                                                                        |
| fileowner()      | It takes filename as an argument and returns the user ID of the owner of specified file.                                                                                                                                                                                                                                                                                                                                                    |
| posix_getpwuid() | It accept user id returned by fileowner() function as an argument and returns an associative array with following<br>References                                                                                                                                                                                                                                                                                                             |

|                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                    |  |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|--|
|                | Keys                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Description                                                                                        |  |
|                | name                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | The shell account user name of the user                                                            |  |
|                | passwd                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | The encrypted user password                                                                        |  |
|                | Uid                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | The ID number of the user                                                                          |  |
|                | Gid                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | The group ID of the user                                                                           |  |
|                | Gecos                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | A comma separated list containing user full name office phone, office number and home phone number |  |
|                | Dir                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | The absolute path to the home directory of the user                                                |  |
|                | Shell                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | The absolute path to the users default shell                                                       |  |
| filegroup()    | It takes filename as an argument and returns the group ID of owner of the specified file                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                    |  |
| posix_getgid() | It accept group ID returned by filegroup() function as an argument and returns an associative array on a group identified by group ID with following refernces                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                    |  |
|                | Key                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Description                                                                                        |  |
|                | Name                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | The name of group                                                                                  |  |
|                | Gid                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | The ID number of group                                                                             |  |
|                | members                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | The number of members belonging to the group                                                       |  |
| filetype()     | It takes filename as an argument and returns the type of specified file . the type of possible values are fifo, char, dir, block, link, file and unknown                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                    |  |
| basename()     | It takes file name as an argument and separate the filename from its directory path.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                    |  |
| copy()         | It takes two string arguments referring to the source and destination file respectively.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                    |  |
| ename()        | It takes two argument as old name and new name and renames the file with new name.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                    |  |
| nlink()        | It takes a single argument referring to the name of file we want to delete.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                    |  |
| is_file()      | It returns true if the given file name refers to a regular file.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                    |  |
| fstat()        | The fstat() function returns information about an open file.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                    |  |
|                | This function returns an array with the following elements:<br>[0] or [dev] - Device number<br>[1] or [ino] - Inode number<br>[2] or [mode] - Inode protection mode<br>[3] or [nlink] - Number of links<br>[4] or [uid] - User ID of owner<br>[5] or [gid] - Group ID of owner<br>[6] or [rdev] - Inode device type<br>[7] or [size] - Size in bytes<br>[8] or [atime] - Last access (as Unix timestamp)<br>[9] or [mtime] - Last modified (as Unix timestamp)<br>[10] or [ctime] - Last inode change (as Unix timestamp)<br>[11] or [blksize] - Blocksize of filesystem IO (if supported)<br>[12] or [blocks] - Number of blocks allocated |                                                                                                    |  |

### Examples

Use of some above mentioned functions is illustrated in the following examples: Example: 1) To read file from server use fread() function. A file pointer can be created to the file and read the content by specifying the size of data to be collected.

```
<?php
$myfile = fopen("somefile.txt", "r") or die("Unable to open file!");
echo fread($myfile,filesize("somefile.txt"));
fclose($myfile);
?>
```

Example: 2) a file can be written by using fwrite() function in php. For this open file in write mode. file can be written only if it has write permission. if the file does not exist then one new file will be created. the file the permissions can be changed.

```
<?php
$filecontent="some text in file";           // store some text to enter inside the file
$file_name="test_file.txt";                 // file name
$fp = fopen ($filename, "w");                // open the file in write mode, if it
does not exist then it will be created.
fwrite ($fp,$filecontent);                  // entering data to the file
fclose ($fp);                               // closing the file pointer
chmod($filename,0777);                      // changing the file permission.
?>
```

Example :

3) A small code for returning a **file-size**.

```
<?php
function dispfilesize($filesize)
{
    if(is_numeric($filesize))
    {
        $decr = 1024; $step = 0;
        $prefix =
            array('Byte','KB','MB','GB','TB','PB');
        while(($filesize / $decr) > 0.9)
        {
            $filesize = $filesize / $decr;
            $step++;
        }
        return round($filesize,2).' '.$prefix[$step];
    }
    else
    {
        return 'NaN';
    }
}
?>
```

### **SetA**

Q1. Write a program to read one file and display the contents of the file with its size.

Q2 Write a program to read a flat file “item.dat”, which contains details of 5 different items such as Item code, Item Name, unit sold, and Rate. Display the Bill in tabular format.

### **SetB**

Q 1. Write a menu driven program to perform various file operations. Accept filename from user.

- a) Display type of file.
- b) Display last access time of file
- c) Display the size of file
- d) Delete the file

**Signature of the instructor:** \_\_\_\_\_ **Date :** \_\_\_\_\_

#### Assignment Evaluation

|              |                      |                     |                      |             |                      |
|--------------|----------------------|---------------------|----------------------|-------------|----------------------|
| 0:Not Done   | <input type="text"/> | 2:Late Complete     | <input type="text"/> | 4:Complete  | <input type="text"/> |
| 1:Incomplete | <input type="text"/> | 3:Needs Improvement | <input type="text"/> | 5: Well Don | <input type="text"/> |